1. INTRODUCTION

Urban bike-sharing systems have emerged as an eco-friendly, cost-effective, and highly accessible mode of public transportation in modern smart cities. These systems mitigate traffic congestion and reduce carbon footprints by providing commuters with an alternative for short-distance travel. However, a major operational challenge faced by bike-sharing providers is maintaining an optimal balance of bicycles across various docking stations. Unpredictable demand fluctuations often lead to situations where popular stations run out of available bikes (shortages), while other stations reach maximum capacity and cannot accept returning bikes (surpluses).

To address this challenge, this project proposes an end-to-end smart city application titled Urban Bike-Sharing Demand Forecasting Using Machine Learning. The primary objective of this system is to accurately predict the future net demand for bicycles at individual stations using historical trip and weather data. By leveraging supervised machine learning algorithms, the system forecasts expected usage patterns and cross-references them against the latest available station status to preemptively identify potential shortages and surpluses.

Furthermore, the application provides an intelligent bike redistribution module. The redistribution engine prioritizes stations with the highest shortages and selects the nearest surplus stations to minimize travel distance and operational cost. Instead of relying on manual observation, the system computes optimal transfer routes, advising fleet operators on exactly how many bicycles to move from surplus stations to shortage stations, complete with distance and estimated time of arrival metrics. The system relies entirely on flat-file CSV data management using Pandas, eliminating the overhead of traditional relational database management systems while maintaining high-performance data processing capabilities.

2. SUPPORTING LITERATURE

2.1 Literature Review

Paper 1 – M. Mergulhão, M. Palma, and C. J. Costa, *A Machine Learning Approach for Shared Bicycle Demand Forecasting*, Proc. 17th Iberian Conf. Information Systems and Technologies (CISTI), 2022.

The paper titled A Machine Learning Approach for Shared Bicycle Demand Forecasting investigates the impact of weather conditions, COVID-19, and pollution on the usage of Bicycle Shared Systems. The authors utilized a dataset sourced from the Gira bike-sharing system in Lisbon, Portugal, covering the year 2020. The dataset included 366 daily records combining bicycle dock status with external factors such as temperature, precipitation, PM2.5 pollution levels, and daily COVID-19 cases. The goal was to establish how these external variables influence daily bike rental demand and to identify the most effective predictive models.

To achieve this, the authors implemented the CRISP-DM methodology. They split the data into 75 percent training and 25 percent testing sets and applied a wide array of machine learning algorithms, including Ordinary Least Squares, Ridge, Lasso, Bayesian Ridge, Support Vector Regression (SVR), Multi-Layer Perceptron Regressor, and Random Forest. Evaluation metrics such as Mean Absolute Error and R-squared score were used to assess model performance.

The results demonstrated that the Random Forest model achieved the highest accuracy, with an R-squared score of 0.920, closely followed by the MLP Regressor. The study concluded that temperature and precipitation are major drivers of demand, while the COVID-19 pandemic did not permanently constrain bicycle shared system usage.

TITLE	A Machine Learning Approach for Shared Bicycle Demand Forecasting
AREA OF WORK	Investigating the impact of weather conditions, COVID-19, and pollution on the usage and demand of Bicycle Shared Systems.
DATASET	Daily records (2020) from the Gira system in Lisbon, combined with temperature, precipitation, pollution, and COVID-19 case data.
METHODOLOGY	CRISP-DM approach with train-test split. Evaluated using MAE, MDAE, MSE, and R-squared metrics through extensive pipeline testing.
ALGORITHM	OLS, Ridge, Lasso, Bayesian Ridge, SVM, MLP Regressor, Random Forest

RESULT/PRECISION	Random Forest achieved the best performance (R-squared = 0.920), followed by MLP Regressor (R-squared = 0.916).
ADVANTAGES	Effectively highlights the non-linear relationship between usage and external factors, proving the resilience of bike-sharing during the pandemic.
FUTURE PROPOSAL	Expanding the analysis over a wider time interval using more complex algorithms to better visualize seasonal fluctuations.

Table 2.1: Summary of Paper 1

Paper 2 – M. Karunanithi, P. Chatasawapreeda, and T. A. Khan, *A Predictive Analytics Approach for Forecasting Bike Rental Demand*, Decision Analytics Journal, vol. 11, 2024.

This paper presents a comprehensive predictive analytics approach to forecasting bike rental demand in the urban environment of Seoul, South Korea. Recognizing that localized surpluses and shortages hinder the efficiency of bike-sharing networks, the authors analyzed extensive historical data encompassing weather variables like temperature, humidity, wind speed, rainfall, snowfall and temporal factors like seasons, holidays, hours of the day. The study involved rigorous Exploratory Data Analysis to uncover the correlations between these factors and bike rental frequency.

For the predictive modeling phase, the dataset was divided into a training and testing split. The study evaluated several advanced machine learning architectures, including Gradient Boosting Regression, Random Forest Regression, Multi-Layer Perceptron Regression, and newer approaches like the Hybrid Autoencoder-Regressor and Transformer models.

The evaluation revealed that the Random Forest Regressor consistently outperformed the other models across multiple metrics, achieving an R-squared of 0.77 and a low Mean Absolute Error of 184.16. The authors concluded that tree-based ensemble methods are highly capable of capturing the intricate, non-linear relationships present in urban mobility data, thereby providing a reliable foundation for strategic bike relocation.

TITLE	A Predictive Analytics Approach for Forecasting Bike Rental Demand
AREA OF WORK	Forecasting bike rental demand in urban areas to enhance operational efficiency and equitable bike distribution.
DATASET	Seoul public bike rental data encompassing temperature, humidity, wind speed, rainfall, snowfall, and holiday indicators.
METHODOLOGY	Extensive EDA, correlation analysis, and a train-test split to train and evaluate multiple advanced machine learning models.
ALGORITHM	OLS, MLP Regression, Gradient Boosting, Random Forest, Hybrid Autoencoder-Regressor, Transformer Model
RESULT/PRECISION	Random Forest was the most accurate, outperforming deep learning and transformer models.
ADVANTAGES	Provides a highly detailed comparison of classical ML models against state-of-the-art neural architectures for bike demand forecasting.
FUTURE PROPOSAL	Implementing artificial intelligence to predict overall demand based on broader urban criteria, such as national and international tourist influx.

Table 2.2: Summary of Paper 2

Paper 3 – C. Gao and Y. Chen, *Using Machine Learning Methods to Predict Demand for Bike Sharing*, Information and Communication Technologies in Tourism 2022, Springer, 2022.

In this study, the authors applied machine learning techniques to predict consumer demand for the Seoul Public Bike sharing system throughout the year 2020. The research distinguishes itself by moving beyond standard weather features to include a highly comprehensive set of 29 features spanning six categories: weather, air pollution, traffic accidents, COVID-19 outbreaks, socioeconomic factors, and seasonality.

The dataset consisted of 9,111 daily observations aggregated from over 2,000 active docking

stations. To isolate the most influential predictors, the authors employed the Boruta feature selection algorithm alongside Recursive Feature Elimination. Four primary machine learning algorithms, namely Linear Regression, K-Nearest Neighbors, Random Forest, and Support Vector Regression (SVR), were then trained using cross-validation.

The results highlighted that the Random Forest and Support Vector Regression (SVR) models delivered the highest predictive accuracy on the testing set. Furthermore, the feature importance analysis confirmed that COVID-19 case counts, air pollution levels, and weather variables like precipitation were the most critical variables driving daily bike rental demand in Seoul.

TITLE	Using Machine Learning Methods to Predict Demand for Bike Sharing
AREA OF WORK	Predicting micro-level consumer demand for bike sharing by incorporating extensive environmental and socioeconomic features.
DATASET	2020 Seoul public bike rentals combined with 29 features across weather, pollution, COVID-19, and socioeconomics.
METHODOLOGY	Boruta and RFE for feature selection, followed by cross-validation and hyperparameter grid search on a train-test split.
ALGORITHM	Linear Regression, K-Nearest Neighbors, Random Forest, Support Vector Regression (SVR)
RESULT/PRECISION	Random Forest achieved the highest testing accuracy, with SVM following closely.
ADVANTAGES	Successfully quantifies the specific impact of unconventional predictors like air pollution and pandemic metrics on bike-sharing demand.
FUTURE PROPOSAL	Modeling short-term, instantaneous demand fluctuations and accounting for station-level geographic variances to optimize bike deployment.

Table 2.3: Summary of Paper 3

2.2 Literature Review Summary

TITLE	SUMMARY
A Machine Learning Approach for Shared Bicycle Demand Forecasting (Mergulhão et al., 2022)	Investigated the impact of weather, pollution, and COVID-19 on Lisbon's bike-sharing system. Found that Random Forest achieved the highest accuracy in capturing these non-linear external factors.
A Predictive Analytics Approach for Forecasting Bike Rental Demand (Karunanithi et al., 2024)	Compared classical ML against deep learning and transformer models for predicting bike demand in Seoul. Concluded that Random Forest Regression provides the most robust and accurate performance.
Using Machine Learning Methods to Predict Demand for Bike Sharing (Gao & Chen, 2022)	Evaluated 29 comprehensive features including socioeconomics and traffic accidents in Seoul. Demonstrated through feature selection that Random Forest is highly effective at modeling diverse environmental predictors.

Table 2.4: Overall Summary of all papers

2.3 Findings and Proposals

The comprehensive review of the three foundational studies clearly highlights the superiority of ensemble machine learning methods, specifically Random Forest, in predicting urban bike-sharing demand. Mergulhão et al. (2022) demonstrated that Random Forest effectively captures the non-linear impacts of daily weather, pollution, and pandemic fluctuations. Karunanithi et al. (2024) further solidified this by proving that Random Forest outperforms even highly complex neural architectures in terms of accuracy and interpretability for this specific domain. Finally, Gao and Chen (2022) showed that tree-based models excel at handling high-dimensional data, efficiently isolating critical variables from broad categories like socioeconomics and air quality.

A consistent limitation across these studies is that while they achieve high predictive accuracy on daily or generalized historical data, they often stop short of translating these predictions into

real-time, actionable redistribution logistics for fleet operators.

Our proposed system bridges this gap. Three machine learning models (Random Forest, XG-Boost, and SVR) are proposed for implementation and comparison. Based on evaluation metrics, including Mean Absolute Error and R-squared, the most suitable model is selected to power the prediction engine. We forecast hourly temporal and meteorological factors to generate precise, station-level predictions of bike deficits and surpluses. More importantly, we advance beyond mere prediction by integrating a dynamic Redistribution Engine. This engine will directly consume the predictions to instantly calculate and recommend optimal, distance-based vehicle routing to resolve localized station shortages and surpluses, prioritizing highest shortage areas and selecting nearest surplus stations to minimize operational cost.

2.4 Problem Statement

Urban bike-sharing systems are critical components of sustainable modern city transportation networks. However, these systems frequently suffer from dynamic imbalances, where sudden surges in usage lead to critical shortages at high-demand stations and capacity overflows at destination stations. Traditional manual relocation methods and static scheduling fail to account for the complex, non-linear influence of weather, time, and special events on commuter behavior. Consequently, there is a pressing need for a predictive system that can accurately forecast station-level demand and provide real-time, actionable redistribution strategies to fleet operators using advanced machine learning techniques.

2.5 Motivation

The increasing adoption of smart city initiatives has created a growing demand for intelligent, data-driven urban mobility solutions. The motivation behind this project is to develop a machine learning-based forecasting system capable of preemptively identifying bike-sharing supply-demand mismatches using historical transit and meteorological data. By integrating predictive modeling with a dynamic redistribution engine, the proposed system not only improves the reliability of public transit but also significantly reduces the operational costs and logistical overhead associated with manual fleet rebalancing. This enhances the overall user experience and supports sustainable urban transportation.

2.6 Research Gap

The literature review indicates that while many existing studies achieve high predictive accuracy in forecasting general daily bike rental demand, they often focus solely on the predictive models themselves without translating those insights into actionable operational tools. Furthermore, several approaches lack sufficient integration of complex, localized environmental factors or do not provide an accessible, real-time deployment platform for fleet operators. To address these gaps, the proposed system integrates advanced ensemble machine learning techniques (Random Forest, XGBoost, and SVR) with a dedicated Redistribution Engine. This engine directly consumes hourly predictions to instantly generate optimal, distance-based vehicle routing recommendations, deployed via an interactive web dashboard for practical, real-world use.

3. SYSTEM ANALYSIS

3.1 Analysis of the dataset

3.1.1 About the dataset

The project utilizes the widely recognized SF Bay Area Bike Share Dataset. This dataset provides a comprehensive, historical record of bike-sharing operations across multiple cities in the San Francisco Bay Area. It is composed of four primary CSV files:

- Station Data: Contains static information about each docking station, including station ID, name, latitude, longitude, and maximum dock capacity.
- Trip Data: A detailed log of individual bike trips, recording start time, end time, origin station, destination station, and journey duration.
- Weather Data: Provides daily meteorological records detailing parameters such as mean temperature, humidity, wind speed, precipitation, and cloud cover.
- Status Data: Provides a massive, minute-by-minute log of bike and dock availability for every station in the network, representing the live status of the system.

3.1.2 Explore the dataset

The merged dataset is specifically designed for predictive modeling of bike demand. Key features include:

- Temporal Features: Hour of the day (0–23), Weekday, Month, Date, Season, and Holiday indicators.
- Meteorological Features: Mean Temperature (°F), Mean Humidity (%), Mean Wind Speed (mph), and Cloud Cover.
- Station Features: Station ID (encoded), Station capacity (dock count), Bike availability, and Dock availability.
- Target Variable: Hourly Net Demand, calculated as the difference between trips originating and trips terminating at a specific station.

3.2 Data Preprocessing

3.2.1 Data Cleaning

The raw CSV files contain inconsistencies and missing values that must be addressed prior to model training. The data cleaning workflow involves:

- Missing Values: Identifying null values in the weather data and applying mean-imputation techniques to ensure continuity across all records.
- Chunk Processing: The status data is processed to generate a continuous hourly time grid for all 70 stations. Advanced feature engineering is then applied exclusively on the chronological training subset to extract historical lag and rolling windows, preventing test-set data leakage.

3.2.2 Analysis of Feature Variables

Feature engineering is performed to extract meaningful predictors for the machine learning algorithms:

- Date Parsing: Timestamps from trip records are parsed using Pandas datetime functions to create cyclic temporal features and holiday/season indicators.

- Data Aggregation: Weather data is aggregated across zip codes to provide a unified daily meteorological record.
- Station Encoding: Station IDs are encoded, and live bike/dock availability metrics are calculated.
- Target Generation: The trip data is grouped by station, date, and hour to precisely compute the target net demand variable.

3.2.3 Analysis of Class Variables

The target class variable is the continuous numerical value representing the Net Demand at a specific station during a specific hour. This is computed by counting the number of trips originating from a station and subtracting the number of trips terminating at that same station. A positive net demand indicates that bikes are leaving the station faster than they are arriving, potentially leading to a shortage.

3.3 Data Visualization

Visualizing the preprocessed data is critical for uncovering underlying mobility trends. The system generates statistical visual representations to understand the true distribution and relationships within the dataset:

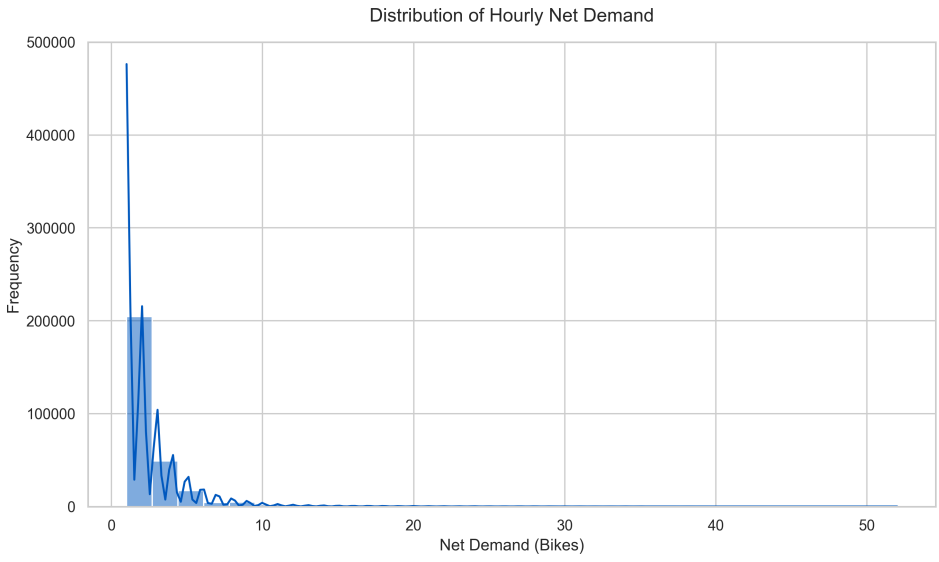


Figure 1: Demand Distribution

The vast majority of hourly records show near-zero or very low net demand, demonstrating that extreme shortages or surpluses are localized outlier events rather than network-wide constants.

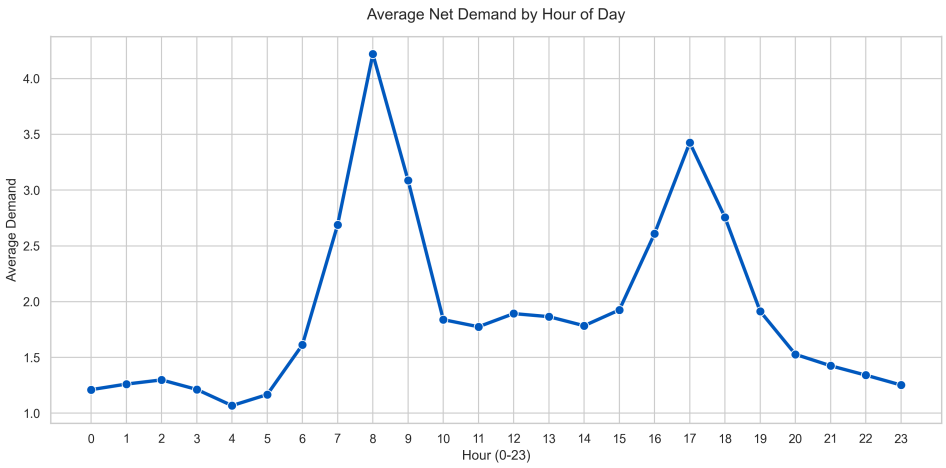


Figure 2: Average Net Demand by Hour

Demand does not rise smoothly; it exhibits sharp, distinct bimodal spikes during the 8:00 AM morning commute and 5:00 PM evening commute windows.

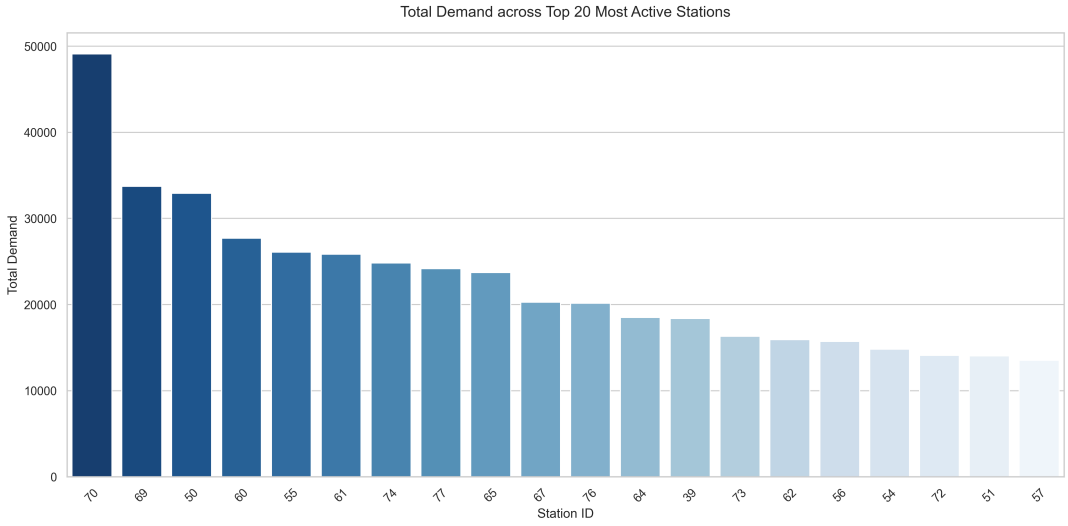


Figure 3: Total Demand by Station (Top 20)

Network activity is heavily skewed. A small subset of specific hub stations dominates total demand, confirming that redistribution efforts can focus on specific high-activity geographic locations.

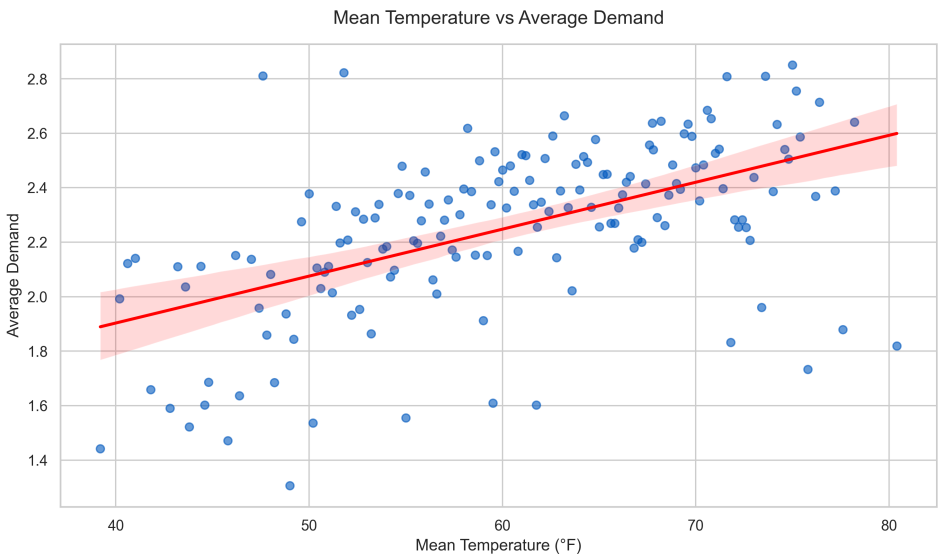


Figure 4: Weather vs Average Demand

There is a slight positive trend indicating higher demand on warmer days, but there is substantial variance at all temperature levels. This indicates that weather alone is insufficient to predict demand without temporal context.

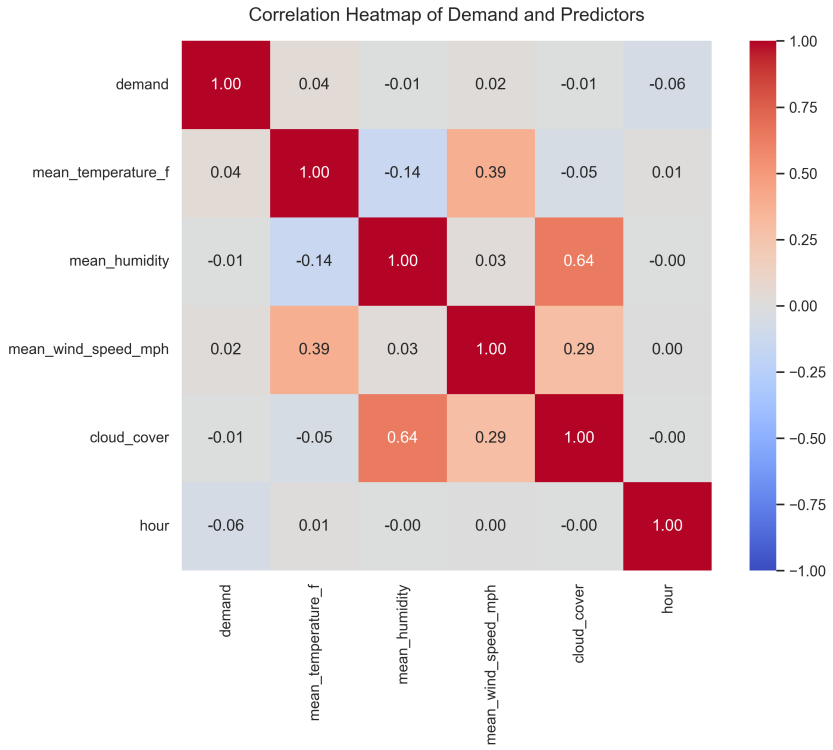


Figure 5: Correlation Heatmap

Raw meteorological features show weak direct linear correlation with raw demand. Temporal variables such as hour provide more useful predictive information, highlighting the importance of time-based patterns in demand forecasting.

3.4 Analysis of Algorithms

3.4.1 Detailed study report of the proposed algorithms

The predictive modeling phase of this project employs three distinct supervised machine learning regression algorithms: Random Forest Regressor, XGBoost Regressor, and Support Vector Regression (SVR). These models were specifically chosen for their proven efficacy in handling complex, non-linear urban mobility data.

Random Forest Regressor

Random Forest is a powerful ensemble learning method that operates by constructing a multitude of decision trees during the training phase. Its basic working principle involves aggregating the predictions of all individual trees to output the mean prediction (regression) of the individual trees, effectively mitigating the risk of overfitting associated with single decision trees. This algorithm was selected for the project because it is highly robust to outliers and exceptionally capable of capturing the non-linear relationships between unpredictable external variables—such as abrupt weather changes and holiday schedules—and bike-sharing demand. Its primary advantage for station-level prediction is its ability to effortlessly handle the high-dimensional feature spaces generated by date and time encoding. However, its main limitation is that the resulting model can be computationally heavy and requires significant memory to store the large ensemble of trees.

XGBoost Regressor

Extreme Gradient Boosting (XGBoost) is an advanced implementation of gradient boosted decision trees designed for speed and performance. Unlike Random Forest, which builds trees independently, XGBoost builds trees sequentially. Each new tree specifically targets and attempts to correct the residual errors made by the previous trees. This iterative boosting process was selected because it often achieves state-of-the-art predictive accuracy on structured tabular datasets. For station-level demand forecasting, XGBoost's primary advantage is its highly optimized execution speed and built-in L1 and L2 regularization techniques, which prevent overfitting

on noisy spatial-temporal data. Its limitation, however, is its sensitivity to hyperparameters; it requires extensive tuning to perform optimally and can be prone to overfitting if the dataset contains extreme anomalies.

Support Vector Regression (SVR)

Support Vector Regression is an extension of the Support Vector Machine classification algorithm adapted for continuous numerical prediction. Its basic working principle relies on mapping the input features into a high-dimensional space using kernel functions, and then identifying a hyperplane that maximizes the margin of tolerance for prediction errors. SVR was included in this project to provide a non-ensemble baseline comparison against the tree-based models. Its main advantage is its theoretical foundation and effectiveness in high-dimensional spaces. The significant limitation of SVR for urban mobility prediction is its poor scalability. It can become computationally expensive and significantly slower to train on the vast, continuous spatial-temporal records generated by the SF Bay Area dataset.

All three models will be rigorously trained using the SF Bay Area Bike Share Dataset. Their predictive capabilities will be independently evaluated and compared against each other using standard regression evaluation metrics, specifically Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared (R^2) Score. The model achieving the best overall performance and the lowest error margin across these metrics will be explicitly selected as the final, operational prediction engine for the real-time bike demand forecasting and redistribution system.

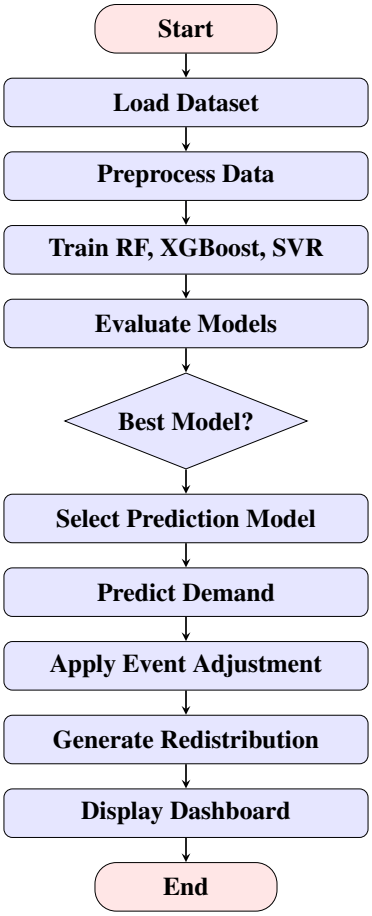


Figure 6: Activity Diagram of the Prediction Algorithm

The step-by-step working of the algorithm is logically structured as follows:

- Load Dataset: The historical dataset is loaded into the system.
- Preprocess Data: The data undergoes cleaning and imputation before the chronological train-test split.
- Train Models: Three machine learning regression models (Random Forest, XGBoost, and SVR) are trained on the prepared dataset.
- Evaluate Models: The models are evaluated and compared using standard metrics.
- Best Model?: The system identifies the model with the highest predictive accuracy.
- Select Prediction Model: The highest performing model is selected for deployment.
- Predict Demand: The selected prediction engine calculates future demand based on user inputs.

- Apply Event Adjustment: Simulated events modify the predicted demand geographically.
- Generate Redistribution: The system prioritizes stations with the highest shortages and selects nearest surpluses to calculate routes.
- Display Dashboard: Interactive KPI cards, dynamic filters, and reports are rendered for the user.

3.5 Project pipeline

3.5.1 Project pipeline diagram with explanation

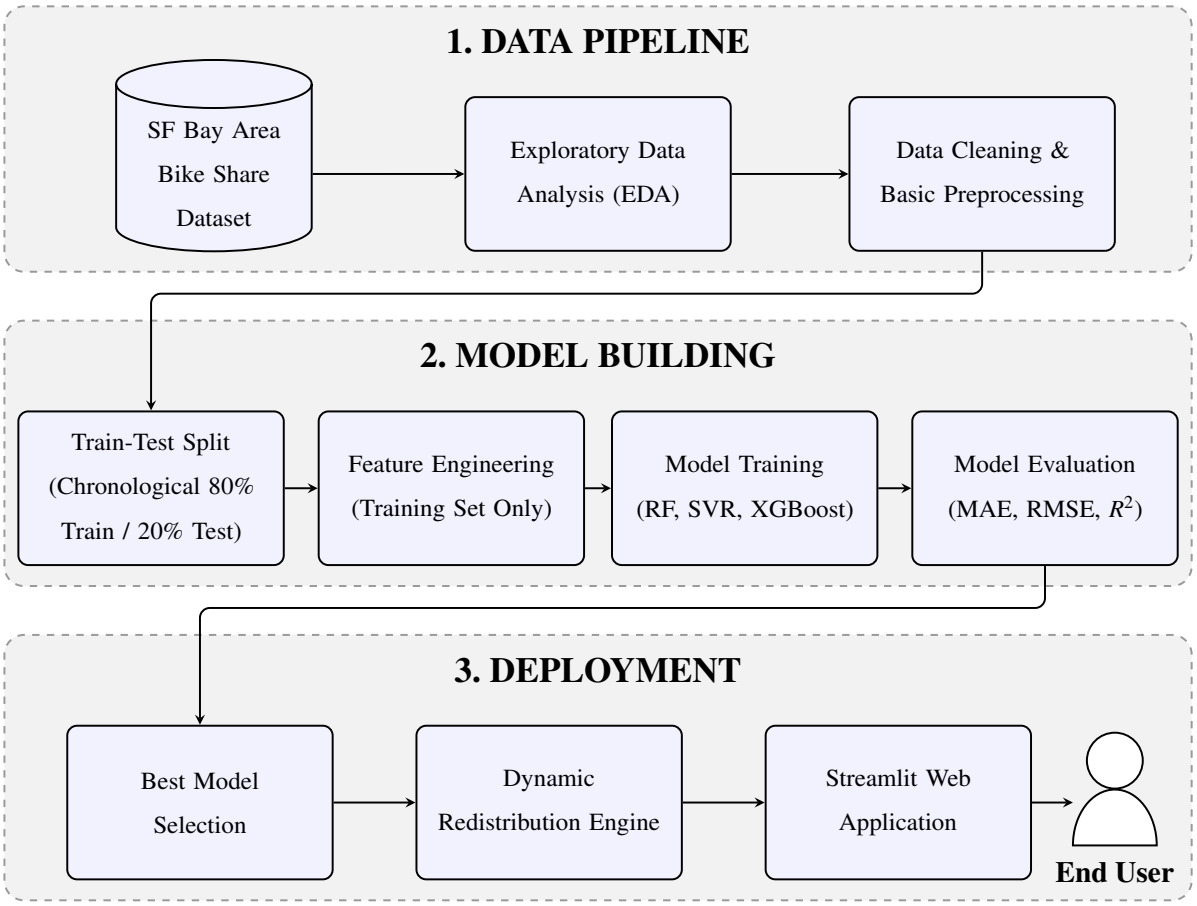


Figure 7: Project workflow

The project pipeline follows a structured three-tier architecture:

- **Data:** The raw dataset is ingested and undergoes Exploratory Data Analysis (EDA) to uncover trends. It is then subjected to rigorous Data Cleaning and Preprocessing to handle missing values, encode variables, and engineer temporal features.

- **Model Building:** The preprocessed data is chronologically split into training and testing sets. Advanced feature engineering is applied exclusively to the training subset. Then, three distinct machine learning algorithms (Random Forest, XGBoost, and SVR) will be trained and evaluated to identify the optimal predictive model.
- **Deployment:** The best-performing model will be deployed through an interactive web dashboard, where end-users can seamlessly interact with the system to generate real-time demand predictions.

3.6 Feasibility Analysis

3.6.1 Technical Feasibility

The system leverages the Python ecosystem, specifically well-documented libraries such as Pandas, Scikit-learn, and Streamlit. The decision to use CSV files and in-memory caching rather than a dedicated database server significantly reduces the architectural complexity and deployment overhead.

3.6.2 Economical Feasibility

The project requires zero financial investment in software licensing or cloud database hosting. All utilized technologies are open-source and free to use. The flat-file data architecture means the system can be hosted locally without incurring costs associated with scaling or API usage limits.

3.6.3 Operational Feasibility

The application provides an intuitive Graphical User Interface. Fleet managers do not need programming knowledge to operate the system; they simply adjust sliders for temperature and time, and the system automatically outputs clear, actionable bike transfer instructions.

3.7 System Environment

3.7.1 Software Environment

- **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu 20.04+)
- **Programming Language:** Python 3.10+

- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, XGBoost, Streamlit, Plotly, Folium
- **IDE/Editor:** Visual Studio Code, Jupyter Notebook
- **Version Control:** Git

3.7.2 Hardware Environment

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher (Multi-core CPU recommended for model training)
- **RAM:** 8 GB minimum (16 GB or more recommended for processing large datasets)
- **Storage:** 10 GB of free disk space (SSD preferred for faster data I/O operations)
- **Network:** Standard broadband internet connection (for package installation and map rendering)